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Optical assembly for optical transceiver and optical transceiver using same

Abstract

Embodiments of the present disclosure are directed to an optical assembly for optical transceiver which is composed of two separate sub-assemblies including an assembly that is coupled with a substrate and has a post formed to have a central hollow into a multi-branched shape, for increasing the optical alignment efficiency between the optical elements included in the optical assembly for optical transceiver which involves multiple complicated and sophisticated processes, the optical assembly for optical transceiver being structured to drain out the epoxy resin or the refractive index matching material used when coupling the optical fiber inserted from the outside with the optical elements, whereby greatly reducing optical alignment errors caused by the epoxy resin or refractive index matching material, as well as directed to an optical transceiver using the optical assembly for optical transceiver.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Ser. No. 16/330,396 filed Oct. 8, 2021, which is a U.S. National Phase Application under 35 U.S.C. 371 of International Application No. PCT/KR2018/013970, filed on Nov. 20, 2018, which claims the benefit of Korean Patent Application No. 10-2018-0002750, filed on Jan. 9, 2018. The entire disclosures of the above applications are incorporated herein by reference. Furthermore, this non-provisional application claims priority in countries, other than the U.S., with the same reason based on the Korean patent application, the entire content of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) Embodiments of the present disclosure relate to an optical assembly for optical transceiver and an optical transceiver using the same.

BACKGROUND

(2) The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

(3) There is a market demand increasing rapidly for transmission of large data files between devices as with data centers, cloud computing, high-performance computing (HPC), ultra-high definition (UHD) and three-dimensional visualization technologies. In addition, the constant increase in market demand for technology of optical interconnections such as rack-to-rack, board-to-board and chip-to-chip interconnections within a system has brought optical interconnect technology to the practical application and commercialization stages.

(4) This trend is leading the increase of bandwidth of the digital interface standard between devices, such as InfiniBand, digital visual interface (DVI), high definition multimedia interface (HDMI), DisplayPort (DP) and USB 3.0. In addition, researches are being actively conducted on small-sized multi-channel optical modules capable of transmitting a large amount of information in order to expand the bandwidth of the digital interface standard between the devices.

(5) In recent years, there has been an increasing demand for board-to-board optical connections in smart devices. Much effort has been devoted to mounting optical modules on smart devices.

(6) However, optical communication technology, which is currently being used commercially, is based on long distance data transmission. Most manufacturers of optical communication components and systems are adapting long-distance optical communication technology to a short-distance optical communication system or a short-distance optical connection. This has been a reason for those inefficient short-distance optical communication systems or short-haul optical interconnect solutions produced by these optical component and system manufacturers.

(7) Therefore, there is a need for a cost-effective optical interconnect solution suitable for short-distance optical communication systems. One of the most cost-effective optical interconnect solutions for high-capacity data transmission and short-haul optical connections is an optical transceiver module using a vertical resonance type semiconductor laser or vertical-cavity surface-emitting laser (VCSEL) and vertical-type photodiodes.

(8) In order to perform optical coupling between an optical fiber and a VCSEL or a vertical-type photodiode, a path of light emitted from the VCSEL or light incident on the vertical-type photodiode should be changed by 90 degrees. An optical system such as a mirror or a prism is required for changing the optical path, and at least one lens is required to improve the optical coupling efficiency.

(9) FIG. **1** is a schematic diagram of an optical system included in a conventional optical transceiver.

(10) The optical system used the conventional optical transceiver includes a transmitter collimator lens **120**, a transmitter reflection prism **130**, a transmitter focusing lens **140**, an optical fiber **150**, a receiver collimator lens **160**, a receiver reflection prism **170** and a receiver focusing lens **180**. Here, the transmitter collimator lens **120**, the transmitter reflection prism **130**, the transmitter focusing

lens **140** are components included in an optical transmitter, and the receiver collimator lens **160**, the receiver reflection prism **170** and the receiver focusing lens **180** are components included in an optical receiver.
(11) The light generated and emitted from a light source **110** has a predetermined radiation angle, is emitted in a direction perpendicular to the surface of the light source **110**, and is incident on the transmitter collimator lens **120**. The transmitter collimator lens **120** converts light incident from the light source **110** into light beams that travel in parallel one another. The transmitter reflection prism **130** changes the path of the light emitted from the transmitter collimator lens **120** by 90 degrees to the side where the optical fiber is. The transmitter focusing lens **140** serves to collect the light reflected by the transmitter reflection prism **130** into the optical fiber **150**.
(12) The light, that is transmitted from the optical fiber **150** and emitted, is incident on the receiver collimator lens **160** which changes the incident light into light beams traveling in parallel one another. As with the optical transmitter, the receiver reflection prism **170** changes and reflects the path of light by 90 degrees. The light reflected by the receiver reflection prism **170** is made incident on a photodiode **190** via the receiver focusing lens **180**, and thereby the optical signal from the optical transmitter finally is delivered to the optical receiver.
(13) The optical transmitter needs a distance equal to the focal length between the transmitter focusing lens **140** and the optical fiber **150** for the purpose of the optical coupling between the optical system and the optical fiber. Likewise, the optical receiver needs a length of optical path for forming parallel light beams between the optical fiber **150** and the receiver collimator lens **160**. Therefore, a special optical fiber alignment mechanism is indispensable for setting these distances.
(14) With such a conventional optical system, precise measuring instruments are necessary because the optical alignment process is required as described above. Additionally, in production of optical transceiver products, optical alignment and assembly performances are sensitive to deviation between the optical system mechanism and the optical fiber mechanism, which requires a very precise management of mechanism deviations.
(15) The conventional optical system exposes the optical fiber **150** in the air, which renders the core of the optical fiber **150** to be susceptible to contamination with fine dust and foreign substances. Depending on the severity of the contamination, the optical coupling efficiency may be fatally influenced. In addition, the exposed core of the optical fiber **150** may generate additional optical coupling loss due to Fresnel loss, resulting in reduced reliability.
(16) In addition, the optical coupling efficiency in the conventional optical system depends on the quality of the cross sectional cut of the optical fiber **150**, which requires special processing of the cross section of the optical fiber **150**. Unless these issues are resolved, products produced using conventional optical systems are highly likely to cause malfunction or failure. Those products, which are commercialized with these inherent deficiencies, cannot be assembled easily by using a complete passive alignment method.
(17) In this way, elaborate arrangements between the light source **110** and the transmitter collimator lens **120**, the transmitter collimator lens **120** and the transmitter reflection prism **130**, the transmitter reflection prism **130** and the transmitter focusing lens **140**, and the transmitter focusing lens **140** and the optical fiber **150** are needed to be controlled within a predetermined level of tolerance to attain high-quality optical transmission. This is also true in the case of optical reception where the light source **110** is replaced with the photodiode **190**.
(18) In other words, in order for the optical transmitter or the optical receiver to operate normally, four individual alignment factors need to be under sophisticated control. Further, time-consuming processes need to be eliminated to allow for mass production of optical assemblies.
(19) Accordingly, a compact optical assembly for optical transceiver is necessary which enables a sophisticated and easy alignment of an optical assembly in optical transceivers for optical communications while allowing a passive alignment of the optical assembly, without requiring expensive equipment or a time-consuming process.

DISCLOSURE

Technical Problem

(20) Embodiments of the present disclosure are directed to providing a compact optical assembly for optical transceiver that can be used in optical transceivers, which enables large capacity optical transmission.

(21) Embodiments of the present disclosure provide an optical transceiver using an optical assembly for optical transceiver, which can be mass-produced at a reasonable price.

SUMMARY

(22) At least one embodiment of the present disclosure provides an optical assembly for use in an optical transceiver and for mounting on a substrate, which includes a body assembly and a cover assembly. The body assembly includes a reflector configured to change a traveling direction of light moving in a z direction perpendicular to a plane formed by the substrate to a −x direction parallel to the substrate, a first lens group disposed between at least one optical element and the reflector to optically couple the at least one optical element and the reflector, a body hole that is an empty space formed at a position spaced apart from the reflector by a predetermined distance, and a surplus liquid guide groove configured to be adjacent to the body hole, and to drain out an overflow of liquid introduced into the body hole. The cover assembly includes a cover spacer inserted in the body hole in a −z direction, and a second lens group arranged on one side of the cover spacer, for optically coupling at least one optical fiber inserted from an outside in an x direction. An optical path is established through an interconnection between the at least one optical element, the first lens group, the reflector, the second lens group, the cover spacer and the at least one optical fiber, by coupling between the cover assembly and the body assembly, with a lower side portion of the cover assembly abutting an upper side portion of the body assembly.

(23) Another embodiment of the present disclosure provides an optical transceiver composed of a substrate, at least one optical element formed on the substrate, a driving circuit for driving the at least one optical element, and at least one optical fiber, and the optical transceiver includes a body assembly and a cover assembly. The body assembly includes a reflector configured to change a traveling direction of light moving in a z direction perpendicular to a plane formed by the substrate to a −x direction parallel to the substrate, a first lens group disposed between the at least one optical element and the reflector to optically couple the at least one optical element and the reflector, a body hole that is an empty space formed at a position spaced apart from the reflector by a predetermined distance, and a surplus liquid guide groove configured to be adjacent to the body hole, and to drain out an overflow of liquid introduced into the body hole. The cover assembly includes a cover spacer inserted in the body hole in a −z direction, and a second lens group arranged on one side of the cover spacer, for optically coupling at least one optical fiber inserted from an outside in an x direction. An optical path is established through an interconnection between the at least one optical element, the first lens group, the reflector, the second lens group, the cover spacer and the at least one optical fiber, by coupling between the cover assembly and the body assembly, with a lower side portion of the cover assembly abutting an upper side portion of the body assembly.

Advantageous Effects

(24) According to at least one embodiment of the present disclosure, an optical assembly for optical transceiver is composed of two separate sub-assemblies including an assembly that is coupled with a substrate and has a bottom post formed to have a central hollow into a multi-branched shape, for increasing the optical alignment efficiency between the optical elements included in the optical assembly for optical transceiver which involves multiple complicated and sophisticated processes.

(25) According to another aspect of the embodiments of the present disclosure, an optical transceiver necessary for large-capacity high-speed optical transmission can be mass-produced at a low cost.

(26) According to yet another aspect of the embodiments of the present disclosure, alignment can

be easily achieved between a plurality of optical elements used in an optical transmitter or an optical receiver without expensive equipment or time-consuming processes.
(27) Yet another aspect of the embodiments of the present disclosure provides an optical transceiver which is structured to drain out the epoxy resin or the refractive index matching material used when coupling the optical fiber inserted from the outside with the optical elements, whereby greatly reducing optical alignment errors caused by the epoxy resin or refractive index matching material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram of an optical system included in a conventional optical transceiver.
(2) FIG. **2** is a schematic diagram of an optical system included in an optical assembly for optical transceiver according to at least one embodiment of the present disclosure.
(3) FIG. **3** is diagrams of an optical assembly for optical transceiver or simply an optical assembly, according to at least one embodiment of the present disclosure.
(4) FIG. **4** is diagrams of a cover assembly of an optical assembly according to at least one embodiment of the present disclosure.
(5) FIG. **5** is diagrams of a body assembly of an optical assembly according to at least one embodiment of the present disclosure.
(6) FIG. **6** is diagrams of a configuration of an optical assembly having a cover assembly and a body assembly coupled together, according to at least one embodiment of the present disclosure.
(7) FIG. **7** is a bottom view of an optical assembly according to at least one embodiment of the present disclosure.
(8) FIG. **8** is a side view of a configuration of an optical assembly coupled to a substrate, according to at least one embodiment of the present disclosure.
(9) FIG. **9** is a view for explaining the principle of coupling an optical assembly to a substrate, according to at least one embodiment of the present disclosure.
(10) TABLE-US-00001 Reference Numeral 110: light source 210: electrical-optical conversion element 120, 160, 220, 262: collimator lens 130, 170, 230, 270, 324: reflector 140, 180, 242, 280: focusing lens 150, 250: optical fiber 190, 290: optical-electrical conversion element 244, 264: spacer 300: optical assembly for optical transceiver 310: cover assembly 310a: first cover groove 310b: second cover groove 310c: third cover groove 310d: fourth cover groove 311: cover spacer 320: body assembly 320a: first body post 320b: second body post 320c: third body post 321, 323: main post 322: reflector forming groove 325, 327: sub-post 328: body encapsulation region 329: optical fiber guide 390: substrate 710: first lengthwise line 720: first widthwise line 730: second lengthwise line 740: second widthwise line 750: third vertical line 1L: first lens group 2L: second lens group G: surplus liquid guide groove
DETAILED DESCRIPTION
(11) Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the following description, like reference numerals designate like elements, although the elements are shown in different drawings. Further, in the following description of some embodiments, a detailed description of known functions and configurations incorporated therein will be omitted for the purpose of clarity and for brevity.
(12) Additionally, various terms such as first, second, i), ii), a), b), etc., are used solely for the purpose of differentiating one component from the other, not to imply or suggest the substances, the order or sequence of the components. Throughout this specification, when a part “includes” or “comprises” a component, the part is meant to further include other components, not excluding thereof unless specifically stated to the contrary.

(13) Hereinafter, an optical assembly for optical transceiver or simply an optical assembly according to some embodiments of the present disclosure will be described with reference to the accompanying drawings. Further, for the sake of clarity, the description will be provided with reference to the Cartesian coordinate system defined in the x direction, y direction and z direction.
(14) FIG. **2** is a schematic diagram of an optical system included in an optical assembly according to at least one embodiment of the present disclosure.
(15) The components of the optical system included in the optical assembly includes a transmitter collimator lens **220**, a transmitter reflector **230** and a transmitter focusing lens unit **240**. Here, the transmitter focusing lens unit **240** includes a transmitter focusing lens **242** and a transmitter spacer **244**.
(16) The transmitter collimator lens **220** changes light beams from an electrical-to-optical signal conversion or electrical-optical conversion element **210** to parallel light beams which are to be transmitted to the transmitter reflector **230**. The transmitter reflector **230** changes the path of the parallel light beams from the transmitter collimator lens **220** by 90 degrees and sends them to the transmitter focusing lens **242**. The transmitter spacer **244** may be configured to have a thickness in the x direction which corresponds to the focal length of the transmitter focusing lens **242**, and thereby allowing the light beams to pass through the transmitter focusing lens **242** to converge on the core of an optical fiber **250**.
(17) The optical assembly according to at least one embodiment of the present disclosure is applicable to the optical receiver side, wherein the collimator lens and the focusing lens swap their functions. Hereinafter, a component including various individual functional elements will be referred to as an assembly.
(18) FIG. **3** is diagram of an optical assembly, according to at least one embodiment of the present disclosure.
(19) An optical assembly for optical transceiver or simply an optical assembly **300**, according to at least one embodiment of the present disclosure includes a cover assembly **310** and a body assembly **320**. A substrate **390**, on which the optical assembly is to be mounted, is formed with four accommodation holes (not shown), while the body assembly **320** is formed with a plurality, specifically a pair, of main posts **321**, **323** and a plurality, specifically a pair, of sub-posts **325**, **327** which are adapted to be directly coupled with the four accommodation holes, and thereby establishing highly reliable and efficient optical alignment between at least one electrical-optical conversion element **210** or at least one optical-electrical conversion element **290** and the body assembly **320**. FIG. **3** does not show the at least one electrical-optical conversion element **210** or at least one optical-electrical conversion element **290**. Hereinafter, the electrical-optical conversion element and the optical-electrical conversion element are commonly referred to as an optical element.
(20) In addition, the optical assembly **300** includes a body encapsulation region **328**. The body encapsulation region **328** serves to completely protect the electrical-optical conversion element, the optical-electrical conversion element, and various other electrical/electronic components and optical components from the outside. As shown in (a) and (b) of FIG. **3**, the body encapsulation region **328** is a space formed by removing a predetermined region inside the lower surface of the body assembly **320** by a predetermined height. The body encapsulation region **328** has a ceiling where a first lens group **1**L is disposed. Thus, by simply coupling the body assembly **320** with the substrate **390**, various components disposed in the body encapsulation region **328** are protected from the outside.
(21) Here, the at least one electrical-optical conversion element **210** or at least one optical-electrical conversion element **290** is disposed on the substrate **390** in a partial area thereof, which is included in the space defined by the body encapsulation region **380**. Here, the first lens group **1**L includes a plurality of lenses arranged in a row.
(22) FIG. **4** is diagrams of a cover assembly of an optical assembly according to at least one

embodiment of the present disclosure.

(23) The cover assembly **310** of the optical assembly according to some embodiments of the present disclosure includes a first cover groove as generally bounded by a dotted rectangle **310***a*, a second cover groove **310***b*, a third cover groove **310***c* and a fourth cover groove **310***d*. The first cover groove **310***a*, the second cover groove **310***b* and the third cover groove **310***c* are coupled with the respective components of the body assembly **320** to be described with reference to FIG. **5**, to assist in the horizontal optical alignment between a second lens group **2**L formed on the body assembly **320** and a reflector **324** formed on the body assembly **320**. Here, the horizontal alignment refers to alignments in the x direction and y direction shown in FIG. **4**. The fourth cover groove **310***d* helps inserting at least one optical fiber **250** into the optical assembly **300**. At least one optical fiber **250** is not shown in FIG. **4**.

(24) In addition, the cover assembly **310** includes a cover spacer **311** as well as the second lens group **2**L. The second lens group **2**L may be formed on one surface of the cover spacer **311**. The second lens group **2**L serves to focus the light beams from the reflector **324** of the body assembly **320** and to transmit the focused beams to at least one optical fiber **250** introduced from the outside. Here, it is assumed that d is the focal length of at least one lens included in the second lens group **2**L. When the thickness of the cover spacer **311**, that is, its length in the x direction is made to be equal to d, the light beams passing through the second lens group **2**L are focused while passing through the cover spacer **311**, and a point on the second interface of the cover spacer **311** becomes the point where the light beams are focused. In other words, the thickness of the cover spacer **311** is the focal length of the second lens group **2**L. Therefore, the light beams passing through the second lens group **2**L are naturally converged on the at least one optical fiber **250** which is in contact with the other surface of the cover spacer **311**.

(25) FIG. **5** is diagrams of a body assembly of an optical assembly according to at least one embodiment of the present disclosure.

(26) In some embodiments, the body assembly **320** of an optical assembly for optical transceiver includes a first body post **320***a*, a second body post **320***b*, a third body post **320***c*, a reflector forming groove **322**, the reflector **324** and a surplus liquid guide groove G. The body assembly **210** also includes the main posts **321** and **323**, the sub-posts **325** and **327**, the body encapsulation region **328**, an optical fiber guide **329** and the first lens group **1**L.

(27) Referring to FIG. **5** at (a), the first body post **320***a*, second body post **320***b* and third body post **320***c* of the body assembly **320** are engaged with the first groove **310***a*, second cover groove **310***b* and third cover groove **310***c* of the cover assembly **310**, respectively, so that the cover assembly **310** can be fixated to be coupled with the body assembly **320**. The first body post **320***a*, when engaged with the first cover groove **310***a*, supports the cover assembly **310** in the −x direction as well as fixates it in the y direction and the −y direction. The second body post **320***b* and the third body post **320***c*, when engaged with the second cover groove **320***a* and the third cover groove **330***a*, respectively, support the cover assembly **310** in the x direction as well as fixate it in the y direction and the −y direction.

(28) The reflector forming groove **322** is formed in a predetermined region of the body assembly **320** to form the reflector **324**. Formed below the reflector **324** is the first lens group **1**L so that it can exchange optical signals with the electrical-optical conversion element or the optical-electrical conversion element disposed in the space protected by the body encapsulation region **328**.

(29) The surplus liquid guide groove G allows a bonding process to be performed by using epoxy, a refractive index matching material or the like for bonding the cover spacer **311** of the cover assembly **310** and at least one optical fiber introduced via the optical fiber guide **329** while they are in contact with each other, in order to block contamination that can be generated on the end face of the at least one optical fiber and to reduce the Fresnel reflection loss by minimizing the refractive index difference between the end face of the at least one optical fiber and the cover spacer **311**. Through this process, the optical assembly according to at least one embodiment of the present

disclosure can maximize optical coupling efficiency.
(30) The pair of main posts **321** and **323** of the body assembly **320** are located on a virtual line formed to pass through the first lens group **1**L, and they perform optical alignment between the first lens group **1**L and the at least one electrical-optical conversion element or the at least one optical-electrical conversion element, either element being disposed on the substrate **390**.
(31) The pair of sub-posts **325** and **237** of the body assembly **320** assist the pair of main posts **321** and **323** when the body assembly **320** is coupled to the substrate **390** to reliably couple the body assembly **320** and the substrate **390** together.
(32) Through the above-described optical alignment structure and method, precise control is achieved between the components of the optical transceiver according to embodiments of the present disclosure. Therefore, cost-effective, precise and easy optical alignment can be provided without time-consuming process, and the resultant optical transceiver can be mass-produced because it is assembled in a fully manual alignment method without expensive equipment.
(33) FIG. **6** is diagrams of a configuration of an optical assembly having a cover assembly and a body assembly coupled together, according to at least one embodiment of the present disclosure.
(34) The cover assembly **310** is coupled to the body assembly **320** to form the optical assembly **300** which is then is mounted on the substrate **390** or other assembly. Coupling the body assembly **320** with the substrate **390** may precede coupling the cover assembly **310** with the body assembly **320**.
(35) Each of the cover assembly **310** and the body assembly **320** may be formed by using a synthetic resin through an injection molding or a three-dimensional printing process.
(36) The substrate **390** may include a rigid printed circuit board made of a rigid material and capable of firmly supporting other components, and/or a flexible printed circuit board made of a soft material and capable of being bent or folded.
(37) The substrate **390** may be mounted with at least one electrical-optical conversion element **210** or at least one optical-electrical conversion element **290**, and a plurality of electric/electronic and optical components for driving thereof. In addition, various components are mounted for large-capacity optical transmission.
(38) FIG. **7** is a bottom view of an optical assembly according to at least one embodiment of the present disclosure.
(39) The optical assembly **300** for optical transceiver, according to at least one embodiment of the present disclosure includes a pair of main posts **321** and **323** and a pair of sub-posts **325** and **327** protruding downward from a lower surface of the body assembly **320**. The pair of main posts **321** and **323** are formed to be symmetric with respect to a first lengthwise line **710** which is a longitudinal line bisecting the lower surface of the body assembly **320**, when they are formed on a first widthwise line **720** which is a virtual line formed to pass through all the lenses included in the first lens group **1**L. Thus, the pair of main posts **321** and **323** are positioned collinear with at least one optical element disposed at a position corresponding to the focal length of the first lens group **1**L, which allows the body assembly **320** to be appropriately adjusted, to thereby adjust the optical alignment between the at least one optical element and the first lens group **1**L.
(40) The pair of sub-posts **325** and **327** to be formed may be positioned as follows. Among other lines extending perpendicular to the first widthwise line **720** virtually passing through all the lenses in the first lens group **1**L, two virtual lines that pass through the pair of main posts **321** and **323** are designated as a second lengthwise line **730** and a third lengthwise line **750**. Formed in parallel with the first widthwise line **720** at a position separated from the first widthwise line **720** by a predetermined distance is a second widthwise line **740**. The pair of sub-posts **325** and **327** are formed on two intersections between the second lengthwise line **730** and the second widthwise line **740**, and between the third lengthwise line **750** and the second widthwise line **740**. The pair of sub-posts **325** and **327** assist the pair of main posts **321** and **323** for allowing the at least one optical element to be optically aligned with the first lens group **1**L when the body assembly **320** is coupled with the substrate **390**.

(41) The pair of main posts **321** and **323** formed to protrude from the lower surface of the body assembly **320** are fixedly inserted in a plurality of main accommodation holes (not shown) formed at positions in the substrate, corresponding to the pair of main posts **321**, **323** when the body assembly **320** is coupled to the substrate **390**. Between the first lens group **1**L of the body assembly **320** and the one or more optical elements disposed on the substrate, the x-direction optical coupling is achieved by side-by-side arrangement of the one or more optical elements on a virtual line that passes through the plurality of main accommodation holes formed in the substrate.
(42) The pair of sub-posts **325** and **327** each has a circular cross-section with the diameter at the lowermost point thereof determined to be smaller than that at its uppermost point, to facilitate inserting the sub-posts in the substrate **390** when they are coupled with the substrate **390**.
(43) Referring to FIG. **7** together with FIG. **4**, the second lens group **2**L to be formed on the cover assembly **310** is positioned with reference to an imaginary straight line bisecting the first cover groove **310***a* in the x direction and to such side that defines the first cover groove **310***a* and extends in parallel with the y direction, among other sides of the first cover groove **310***a*. Further providing the cover spacer **311** with an appropriate thickness in the x direction, adjusted to the focal length of each lens included in the second lens group **2**L establishes the optical coupling from the reflector **324** to the second lens group **2**L and the optical coupling from the second lens group **2**L via the cover spacer **311** to the at least one optical fiber **250**. Accordingly, coupling the cover assembly **310** to the body assembly **320** completes the optical coupling established through an interconnection between the at least one optical element, the first lens group **1**L, the reflector **324**, the second lens group **2**L and the at least one optical fiber **250**.
(44) FIG. **8** is a side view of a configuration of an optical assembly coupled to a substrate, according to at least one embodiment of the present disclosure.
(45) The optical assembly **300** according to at least one embodiment of the present disclosure is fastened to the substrate **390** by inserting the pair of main posts **321**, **323** formed to protrude from the lower surface of the body assembly **320** in the main accommodation holes formed in the substrate.
(46) The cover assembly **310** and the body assembly **320** when coupled together, form the optical assembly **300** which is then mounted on the substrate **390** or other assembly. Coupling the cover assembly **310** to the body assembly **320** may come after coupling the body assembly **320** with the substrate **390**.
(47) The substrate **390** may be mounted with one or more electrical-optical conversion elements **210** or one or more optical-electrical conversion elements **290**, and a plurality of electric/electronic and optical components for driving thereof. In addition, various components for large-capacity optical transmission are mounted.
(48) The electrical-optical conversion elements **210** or optical-electrical conversion elements **290** and a plurality of electric, electronic and optical components for driving thereof are arranged on the substrate **390**, to which the optical assembly **300** are coupled, to finally complete an optical transceiver.
(49) FIG. **9** is a view for explaining the principle of coupling an optical assembly to a substrate, according to at least one embodiment of the present disclosure.
(50) FIG. **9** shows at (a) a side view of the substrate **390**. The substrate **390** may be formed to have a plurality of accommodation holes that become narrower from the top to the bottom.
(51) FIG. **9** shows at (b) and (c) a side view and a bottom view of a pair of main posts **321** and **323**, respectively. The pair of main posts **321**, **323** are formed protruding from the lower surface of the body assembly **320**.
(52) Each of the pair of main posts **321** and **323** has a hollowed-out cylinder composed of a first cylinder having a first diameter and a hollow formed by a concentric second cylinder having a second diameter that is smaller than the first diameter. Then, at least two openings are formed through the first cylinder to communicate between outer circumferential surfaces of the first

cylinder and the center of the first cylinder so that the hollow is exposed externally in radial directions from the center. Here, the openings may be formed at predetermined angles along the outer circumferential surfaces of the first cylinder having the hollow. For example, as shown in FIG. **9**, when the openings are each formed at every 90 degrees along the outer circumference, the first cylinder including the hollow will be divided into four columns. When the openings are each formed at every 60 degrees along the outer circumference, the first cylinder including the hollow will be provided with six divided columns.

(53) With the hollows and openings formed respectively in the pair of main posts **321** and **323**, **390**, coupling the body assembly **320** to the substrate **390**, that is, inserting the pair of main posts **321** and **323** in the accommodation holes formed on the substrate **390** causes the divided columns respectively constituting the pair of main posts **321**, **323** to yield under a force toward the center of the divided columns, until they enter the accommodation holes. In order to utilize this property, the body assembly **320** including the pair of main posts **321** and **323** is preferably formed of a synthetic resin having a certain degree of elasticity. However, the material of the body assembly **320** is not limited thereto. With the plurality of main posts **321**, **323** configured to have such feature, the precise optical alignment is achieved between at least one electrical-optical conversion element **210** or optical-electrical conversion element **290** disposed on the substrate **390** and the optical assembly **300** for optical transceiver.

(54) On the other hand, the surface on which the regions where the pair of main posts **321** and **323** are protruded is formed to have stepped portions with a predetermined depth elevated above the contact surface of the body assembly **320** when engaged with the substrate **390**. Thus, even when the optical assembly **300** is coupled to the substrate **390**, the body assembly **320** maintains non-contact regions formed on at least the surface thereof where it forms a border with the pair of main posts **321** and **323**. The non-contact region is called an anti-delamination region.

(55) When the optical assembly **300** is coupled to the substrate **390**, the pair of main posts **321** and **323** may be deformed, which could cause the lower surface of the body assembly **320** adjacent to the main posts **321**, **323** to be separated from the substrate **390**. Such a phenomenon will be eliminated by the formation of anti-delamination region.

(56) Although exemplary embodiments of the present disclosure have been described for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the idea and scope of the claimed invention. Therefore, exemplary embodiments of the present disclosure have been described for the sake of brevity and clarity. The scope of the technical idea of the present embodiments is not limited by the illustrations. Accordingly, one of ordinary skill would understand the scope of the claimed invention is not to be limited by the above explicitly described embodiments but by the claims and equivalents thereof.

Claims

1. An optical assembly for use with an optical transceiver and for mounting on a substrate, the optical assembly comprising: a body assembly comprising: a reflector configured to change a traveling direction of light moving in a z direction perpendicular to a plane formed by the substrate to a −x direction parallel to the substrate, a first lens group disposed between at least one optical element and the reflector to optically couple the at least one optical element and the reflector, a body hole that is an empty space formed at a position spaced apart from the reflector by a predetermined distance, a surplus liquid guide groove configured to be adjacent to the body hole, to be elongated in a y direction or a −y direction, penetrating a side of the body assembly, and to drain out an overflow of liquid introduced into the body hole only toward the side of the body assembly, wherein the bottom of the surplus liquid guide groove is configured to be higher than the bottom of the body hole; and a plurality of main posts formed on a first widthwise line that is a virtual line

formed to pass through all lenses included in the first lens group, and at least two sub-posts formed on a first lengthwise line and a second lengthwise line that are two virtual lines formed perpendicular to the first widthwise line, wherein one of the sub-posts is formed on the first lengthwise line and another of the sub-posts is formed on the second lengthwise line; and a cover assembly comprising: a cover spacer inserted in the body hole in a −z direction, and a second lens group arranged on one side of the cover spacer, for optically coupling at least one optical fiber inserted from an outside in an x direction, wherein an optical path is established seamlessly from the at least one optical element to the at least one optical fiber through the first lens group, the reflector, the second lens group and the cover spacer in order, by coupling between the cover assembly and the body assembly, with a lower side portion of the cover assembly abutting an upper side portion of the body assembly, and wherein each of the plurality of main posts comprises: a combination of a plurality of columns collectively having a cross-sectional shape formed by a ring-shaped figure which is composed of a first circle having a first diameter, and a circular hollow formed by a concentric circle having a second diameter that is smaller than the first diameter, and further includes at least one rectangular hollow having a first pair of opposite sides that are equal to the first diameter and a second pair of opposite sides that are smaller than the first diameter, to allow the body assembly to be repositioned in all lateral directions in parallel with the substrate when the body assembly is coupled to the substrate, whereby allowing one or more of the optical elements to be each optically coupled with each lens included in the first lens group.

2. The optical assembly of claim 1, wherein the body assembly comprises: a first body post, a second body post and a third body post formed at positions corresponding to a first cover groove, a second cover groove and a third cover groove, respectively, to facilitate the coupling between the cover assembly and the body assembly.

3. The optical assembly of claim 2, wherein the first body post is configured to fix the first cover groove in the y direction and the −y direction as well as support the first cover groove in the −x direction, and the second body post and the third body post are configured to support the second cover groove and the third cover groove in the y direction and the −y direction, respectively and in the x direction commonly, for allowing the cover assembly to be tightly coupled with the body assembly when the cover assembly is coupled with the body assembly.

4. The optical assembly of claim 3, wherein one or more of the optical elements are each optically coupled with each lens included in the first lens group by a plurality of main accommodation holes formed at positions in the substrate, corresponding to the plurality of main posts, and the at least one optical element disposed between the main accommodation holes, by coupling between the body assembly and the substrate.

5. The optical assembly of claim 4, wherein the at least two sub-posts are configured to assist the plurality of main posts for allowing one or more of the optical elements to be each optically coupled with each lens included in the first lens group, when the body assembly is coupled with the substrate.

6. The optical assembly of claim 5, wherein the second lens group is formed based on a virtual straight line bisecting the first cover groove in the x direction, the first vertical line, the second vertical line and a thickness of the cover spacer along the x-direction, so that coupling between the cover assembly and the body assembly establishes an seamless optical coupling from the at least one optical element to the at least one optical fiber through the first lens group, the reflector and the second lens group in order.

7. The optical assembly of claim 6, wherein each of the plurality of main posts comprises: a hollowed-out cylinder composed of a first cylinder having a first diameter and a hollow formed by a concentric second cylinder having a second diameter that is smaller than the first diameter, and at least two openings formed through the first cylinder at predetermined angles to communicate between outer circumferential surfaces of the first cylinder and a center of the first cylinder so that the hollow is exposed externally in radial directions from the center.

8. The optical assembly of claim 6, wherein the body assembly has a lower surface comprising: an anti-delamination region formed by stepping the body assembly at least partially where the plurality of main posts protrude from, to prevent the body assembly from lifting from the substrate when the body assembly is pressurized locally at junction between at least one of the plurality of main posts under a pressure in a horizontal direction parallel to the substrate while the body assembly is coupled to the substrate.